

PM-003 Hardware Selection

Project: Mochi V1.0

Status: Revision A (Draft)

The components below are the current recommended hardware for Project Mochi based on the design goals and budget.

Hardware List

Component	Selected Part	Reason
Main Computer	Luckfox Pico Plus	Linux, Python support, compact, budget-friendly
Display	1.54-inch IPS LCD (240×240)	Expressive face, good size
Camera	OV5647 CSI Camera	Compatible with Luckfox
Drive Motors	2× N20 Metal Gear Motor, 6V, 200 RPM	Reliable, compact
Head Servo	small geared DC motor with an encoder	Strong and durable for head tilt
Motor Driver	DRV8833	Drives two DC motors efficiently
Battery	1000 mAh Flat LiPo	Compact and lightweight
Speaker	28 mm 8Ω Speaker	Good sound for size
Microphone	I ² S Microphone Module	Voice input
Charging	USB-C LiPo Charging Module + Dock	CS supports future auto docking
Wheels	40 mm Rubber Wheels	Good grip on desks
Caster	Mini Ball Caster	Stable third contact point
Fasteners	M2 & M2.5 Screws	Standard hardware

Estimated Budget: Approximately US\$80–90, leaving room for unexpected costs while staying below the US\$100 target.